FishAI v2.0: The Three-Sided Ask
Refuting an Avoidance Rule and Shipping Its Inverse

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Abstract

Every ask in Literature (Canadian Fish) is three things at once: a bet on a card, a *broadcast* — the holding licence of `us54` row 6 makes an ask publish that the asker holds a card of the named set — and a *concession*, because row 10 hands the turn to whoever was asked on a miss. The engine has only ever scored the first. The project's owner asked for "off-limits" reasoning: identify the opponents who would punish a conceded turn, and refuse to ask them. Implemented as a penalty on the miss branch and measured on duplicate deals against a control that reproduces the shipped decision function byte-for-byte, it **loses** 4.5 to 11.7 points of win rate, monotonically worse as the appetite rises (-0.52 to -1.61 sets per duplicate pair), while an information-free perturbation of the same magnitude is free (-0.12 ± 0.44 at matched size). The cause is structural and is this paper's central idea: row 6 makes threat and opportunity *the same fact*. A seat is dangerous exactly when it holds a card of a set this team is invested in, which is what makes it the seat most likely to hold the card this team wants — $\mathrm{corr}(\mathrm{threat}, \mathrm{best\ available\ hit\ probability}) = +0.249$ over 33,595 observations, mean best p of 0.471 at high-threat seats against 0.326 elsewhere. The sound response is therefore not avoidance but **defusal**: ask *into* the set the dangerous seat has published a basis in, take the card its reach rests on, and keep the turn under row 9. Same threat model, opposite prescription, and it wins: $+1.43$ and $+1.58 \pm 0.32$ sets per duplicate pair on two held-out banks, positive at all nine roster styles, $+1.08$ to $+1.96$ against opponents that also defuse, and — the validity result that matters most — $+1.55$ to $+1.75$ inside a third party's engine against their bots, with one knob moved. Most of the paper is negative. Signalling is flat and its one positive arm failed to replicate (-0.013 ± 0.033 at 1,500 pairs); stalling is inert when weak and -1.14 ± 0.19 sets when driven hard; the turn-handoff request is built and refuted, the shipped hand-size rule already picking the best teammate 84.8% of the time with 73.1% of those decisions carrying no signal at all; and a contained-pass aim fix measured below the instrument's floor and was *reverted rather than shipped*, because a listed-but-inert knob contaminates a measured vector. Concealment is a counter-weapon worth $+0.97 \pm 0.27$ added to defusal against defusing opponents, and *worse than plain when played alone.* This project's own intransitive-triangle claim is refuted at 4,000 pairs per edge on two independent bank sets — the 2×2 is a strict total order $DC > D > P > C$ — and replaced by an opponent-dependent argmax worth one bit and up to $+0.29 \pm 0.14$. Two engine defects are reported: `missTarget: 'fewest'` has the wrong sign (hand size correlates -0.147 with the cards a conceded seat takes, against the licence estimator's $+0.421$), and `style.signalling` is unreachable under `us54`. Two facts are stated rather than buried: the certain-hit gate exists because an adversarial pre-commit review found the defusal credit could displace a certain hit (9 occurrences over 150 games $\times$ 9 styles against 0 for the control), and the project's own detection-floor characterisation reclassifies two of this paper's nulls as non-measurements rather than zeros.

1 Introduction

An ask in Literature is a question, and the engine has always treated it as one: rank the legal asks by the probability that the named card is where you think it is, filter the long shots, break the ties, play the argmax [11]. That model is correct as far as it goes, and it goes about a third of the way. Under the `us54` dialect [7] the same ask does two other things, both consequential, neither scored. It **broadcasts**: row 6 permits an ask only into a set the asker holds a card of and row 7 forbids naming a card you hold, so every ask publishes that the asker holds a card of this set and lacks this one — the only reliable public fact about a hidden hand in this game, and what the engine's whole inference layer is built on [14, 10]. And it **concedes**: row 9 keeps the turn on a hit, row 10 hands it to the seat that was asked on a miss, so the miss branch of every ask carries a cost that depends on *which* seat receives the turn — a cost the engine has priced by hand size alone.

The work began with a request from the project's owner, phrased as *off-limits* reasoning: build a model of which opponents would punish a conceded turn, and refuse to ask them. The threat model that request describes is sound, derivable from the public log, and — the first surprise — a strict improvement on what the engine already uses for the same job (Section 4). The prescription is not sound. Penalising asks by the danger of the seat they would concede to costs up to 11.7 points of win rate, and an information-free perturbation of the same size costs nothing, so the loss is caused by the threat *information* rather than by the disturbance (Section 5).

The reason is the paper's organising idea, and it fits in a sentence: **row 6 makes threat and opportunity the same fact**. A seat is dangerous precisely when it holds a card of a set this team is heavily invested in, and that is precisely what makes it the seat most likely to be holding the card this team wants (Section 5.2), so avoiding it means abandoning your own best set.

So keep the model and invert the prescription: ask the dangerous seat — **defuse**, taking back the very card its reach rests on while row 9 keeps the turn. That mechanism ships, and Section 6 measures it at increasingly hostile scales, ending inside a third party's engine against a third party's bots (Section 7). The rest is negative and deliberately so: Sections 8 to 11 report three further requested mechanisms refuted and a fourth that works only as a counter to the first, Section 12 retracts a structural claim this project published about its own tactic space, and Section 13 reports what the instrument underneath all of it can resolve [17].

2 Related work and prior results in this series

The measurement discipline is the project's own and is unchanged here: duplicate deals (every seeded deal played in both orientations, so the deal is never a confound), the pair as the unit of analysis, paired standard errors, and byte-exact controls wherever a mechanism can be switched off [11, 9]. Four prior results are load-bearing: the v0.5 roster's nine styles and tuned Balanced control [11, 8]; the v1.0 counter table whose dominant row makes style-switching provably degenerate, which Section 15 prices the per-decision layer against [12, 13, 6]; the contained-book paper's value of a turn, $E = \text{hits}/\max(1, \text{misses})$, and its independent estimate of the cards a conceded turn is worth, which Section 4.1 agrees with to within a few hundredths [16, 4]; and the detection-floor characterisation that supplies Section 13's resolution limits [17]. The external anchor is the **fishlabs** repository [19], an independent Canadian Fish project with its own engine, its own bots, and — usefully — an independently invented implementation of the very rule Section 5 refutes.

3 Background: the rules the results rest on

Six players in two teams of three (seats $0, 2, 4$ against $1, 3, 5$); 54 cards in nine half-suits ("sets") of six; nine cards dealt to each seat; the game ends the moment a team has been awarded five sets, so ties are arithmetically impossible [7, 20]. The rows this paper cites by number [7]:

row	rule
6	**Ask licence.** The asker must hold at least one card of the asked set.
7	**Own card.** You may not ask for a card you hold.
8	**Target has cards.** The target must hold at least one card.
9	**Hit.** The card transfers; the asker *keeps the turn* and asks again.
10	**Miss.** The turn passes to the player who was asked.
11	**Declare timing.** Any time, including out of turn, inside the declare window.
14	**Any declare error at all.** The opposing team scores the set.
17	**Information.** Counts are public, with a full log of asks, results and declares.

Rows 6 and 7 make the broadcast; rows 9 and 10 make the concession. Row 11 matters twice below: it is why the value of a turn is the value of *asks* rather than declares, and why the turn-handoff request loses a premise (Section 10).

4 What a conceded turn actually costs

4.1 The instrument

`scripts/probe-danger.mjs` records a concession whenever an ask is about to miss. The outcome variable is how many cards the conceded seat then takes from the conceding team before it misses in turn — the length of row 9's hit chain, in cards — read from the hands: this is a measurement harness, not a bot. The *estimate* it is compared against is built by `buildKnowledge` over the acting seat's `SeatView`, i.e. exactly the information a policy has [1, 10]. Over 300 `us54` games, all-Balanced seats, 12,376 conceded turns: the mean cards taken per conceded turn is **1.304**, with $P(0) = 46.7\%$, $P(\geq 2) = 33.2\%$ and a maximum of 11. That mean lands within a few hundredths of the contained-book paper's independent derivation of the same quantity from E [16, 4], and two arguments from different premises agreeing is the reason to trust either.

4.2 Two estimators, one of which is backwards

- `handSize` — `view.counts[X]`. What the engine uses today, in two places: `missTarget` in `decide.ts`'s `pickAsk`, and `valueContainedPass`'s $E \cdot n_t/\bar{n}$ term [10, 16].
- `licence` — over the sets X has publicly shown a basis in (row 6, read off the log), the cards of those sets that this seat can certainly locate on its *own* team. X's reachable prey.

Over the 12,376 observed concessions, `handSize` correlates $-\mathbf{0.147}$ with the cards the conceded seat then took, and `licence` correlates $+\mathbf{0.421}$, with a 3.2× lift from its bottom decile (0.778 cards) to its top (2.469) [1]. That is the first finding of the work, and it is a defect report: the shipped proxy has the wrong sign. `missTarget: 'fewest'` prefers to concede to the opponent with the smallest hand, on the stated grounds that the surrendered turn is worth least there; measured, the smallest hand takes *more*. Hand sizes fall as information accumulates, so the sign could have been a phase artifact. It is not: over log lengths 0–40, 40–80 and 80–120 (n = 5,216/4,626/2,452, mean cards taken 1.19/1.37/1.40), corr(`hand`) runs $-0.191/-0.149/-0.100$ against corr(`licence`)

Table 1: The owner's off-limits rule swept over appetite: a danger penalty on the miss branch, 200 duplicate pairs per cell [1]. The first column is the byte-exact control.

appetite	0 (control)	0.05	0.10	0.25	0.50	1.00
win rate	50.00%	45.0%	43.0%	45.5%	42.3%	**38.3%**
paired set-difference	0.0000 ± 0.0000	−0.52	−0.73	−0.89	−1.19	**−1.61**

at $+0.474/+0.416/+0.353$ — negative in every bucket, and Section 10's independent fork study reproduces the sign a third time at −0.261. Least squares over the licence estimator gives the coefficients the shipped `threat.ts` carries:

$$\text{cards} \;\approx\; 0.885 \;+\; 0.391 \cdot \text{prey}. \tag{1}$$

4.3 Where the licence must be read from

Not from `Knowledge.constraints`. The knowledge engine records an ask as a deal-time set-constraint and *drops it the moment it is satisfied or exhausted* — which is exactly when the seat has been shown to hold a card of the set, i.e. when the threat is most real. Reading the basis off the public log instead multiplies what is visible, though not by the same factor twice: detection of a harvest threat (opponent basis plus at least two of the set on my team) roughly doubles, from 6.0% to **11.7**%, while detection of the owner's strict five-and-one position rises about fivefold, from 7.1% to **35.0**% [1]. Those figures are also why every mechanism here is a **graded term and never a veto**: even at its best the predicate is invisible about two thirds of the time, and a rule that fires on a third of its cases and is silent on the rest cannot be a hard prohibition without being arbitrary.

5 Off-limits, as requested: refuted

The request, implemented literally, is a penalty on the miss branch,

$$\text{penalty} \;=\; w_{\text{hit}} \cdot (1-p) \cdot a \cdot \frac{D(t)}{1+E}, \tag{2}$$

subtracted from the shipped ask score, where p is the seat's own hit estimate, $D(t)$ the threat estimate of Equation (1) at target t, E the value of a turn in cards, and a the appetite. Two hundred duplicate pairs per cell, against a prototype control that reproduces `decide` byte-for-byte when the mechanism is off [1].

Table 1 is monotone in the wrong direction: the harder the rule is driven, the worse it does, from −0.52 to −1.61 sets per duplicate pair and 4.5 to 11.7 points of win rate below the control. A hard veto restricted to the narrow case — likely-miss asks only, at targets with at least K certainly-located prey cards — loses too, at −1.02 to −1.38 for $K = 2 \ldots 5$, reaching neutral only at $K = 6$, where it has stopped firing.

5.1 The null control, which is what makes this a finding

A policy perturbation that costs win rate has two candidate causes: the information it uses, or the mere fact of disturbance. The separating experiment is the same penalty at the same magnitude with D replaced by an information-free hash of the target. At appetites 0.10/0.25/0.50 the null measures 48.50%/50.25%/50.00% win rate and -0.12 ± 0.44, $+0.01 \pm 0.42$, -0.23 ± 0.44 sets per pair [1]. **An information-free perturbation of the same size is free**, and the danger-informed one costs 4.8 to 7.7 points more at matched magnitude (48.5 against 43.0; 50.25 against 45.5; 50.0 against 42.3). The loss is caused by the threat information, not by the noise.

5.2 Why: threat and opportunity are the same fact

Row 6 cuts both ways, and here is what that means numerically. Over 33,595 ask decisions, $\mathrm{corr}(\mathrm{threat}(t), \text{best hit probability available at } t) = +\mathbf{0.249}$: the 12,691 high-threat targets (at least two reachable prey cards) carry a mean best available p of $\mathbf{0.471}$ against $\mathbf{0.326}$ at the 20,904 low-threat ones [1]. A seat is dangerous *because* it holds a card of a set this team is heavily invested in — and that is the same fact that makes it the seat most likely to be holding the card this team wants. Penalising threat is penalising your own best set.

There is also a soundness argument that reaches the same conclusion without any win rate, and it generalises past this engine. A threat model built from public information is a strict **under-approximation** of what a seat knows. A positive fact in it — *this seat has published a basis in that set* — is a certificate and is sound. An absence is not evidence: the seat may hold a basis it has never had occasion to publish. So the model may legitimately be used to act on *proven* reach, and may **never** be used to certify a seat as safe. Avoidance requires exactly the forbidden direction; defusal requires only the sound one.

6 Defusal: the same model, the opposite prescription

If the dangerous seat is the seat holding the card you want, the answer is not to leave it alone — it is to take the card back. A hit removes the very card the licence rests on, and row 9 keeps the turn while doing it. The mechanism is therefore a bonus on the *hit* branch where the off-limits rule was a penalty on the *miss* branch:

$$\text{bonus} = \begin{cases} a_d \cdot w_{\text{hit}} \cdot p \cdot \dfrac{c \cdot \mathrm{prey}(B)}{1+E} & \text{if the target has a published basis in } B, \\ 0 & \text{otherwise,} \end{cases} \tag{3}$$

added to the ranked ask score, where B is the asked card's set, $\mathrm{prey}(B)$ the number of cards of B certainly located on this team, c the fitted cards-per-prey-card slope of Equation (1), and a_d the style's appetite [10]. Dividing by $1+E$ converts the credit from cards into the ranker's own units, so the appetite means the same thing at every hit rate.

Two design decisions were measured rather than assumed. **Per set, not per seat**: crediting the target's whole published reach was measured too, and per-set is both better (+1.50 against +1.21 on the same bank) and the only version whose causal story is true, since a hit on a card of B cannot protect a set it is not in. **A term, not a branch**: because the credit is added to the score, two of `pickAsk`'s three guarantees survive for free — `minHitP` still filters the same pool, and both near-tie windows still break the same ties. The third did not.

6.1 The certain-hit gate, and why it exists

The gate was not designed in. It was added because an adversarial pre-commit review found that the credit could displace a certain hit, and then measured how often. The arithmetic is thin in the wrong direction: a certain hit's score margin over an uncertain ask is about 2 points, while Equation (3) is bounded only by $a_d \cdot w_{\text{hit}} \cdot c \cdot \mathrm{prey}/(1+E)$, which reaches about 137 at $w_{\text{hit}} = 70$. Ungated, the term abandoned a certain hit **9 times over 150 games $\times$ 9 styles, against 0 for the control** [1, 10].

Table 2: Defusal, shipped implementation: each roster style against itself with `defuse: 0`, 400 duplicate pairs per cell [1]. Intervals are ± 1.96 paired SE.

cell	win rate	paired set-difference
balanced, holdout A	60.88%	$+\mathbf{1.43 \pm 0.32}$
balanced, holdout B	61.38%	$+\mathbf{1.58 \pm 0.32}$
blitz	61.88%	$+1.35 \pm 0.32$
punter	58.38%	$+1.22 \pm 0.33$
banker	57.63%	$+1.12 \pm 0.31$
turtle	61.25%	$+1.35 \pm 0.36$
hoarder	59.75%	$+1.31 \pm 0.35$
scout	61.63%	$+1.66 \pm 0.32$
ghost	55.38%	$+0.83 \pm 0.32$
archivist	**65.00%**	$+\mathbf{1.94 \pm 0.33}$

`pickAsk` therefore zeroes *both* concession terms for any ask with $p < 1$ whenever a certain hit is available, leaving the credit live *among* certain hits, where it is free. Section 6.2's table was re-measured with the gate in place and did not move. Two honest riders travel with it, both in Section 17: its firing rate is unpriced beyond the sweep that found it, and its concealment half is not pinned by a test.

6.2 Measured, on held-out banks and across the roster

The appetite was fixed at 1 on a tuning bank — the sweep is an inverted U peaking at 1–2 and back to noise by 8 — then confirmed on two disjoint held-out banks and across the roster. Table 2 is the *shipped* implementation, not the prototype: `STYLE_ROSTER[s]` against the same style with `defuse: 0`, 400 duplicate pairs per cell [1].

Every interval excludes zero; the sign replicates on both banks and at all nine styles.

Two sets of numbers appear in this literature and they are not the same measurement. The prototype — the term alone, hooked into a mirror of `pickAsk` — measured $+1.50$ $[1.24, 1.75]$ and $+1.65$ $[1.38, 1.91]$ on 600-pair held-out banks, and those are the figures `defuse.ts` carries in its header as the result it was built from. Table 2 is the shipped implementation re-measured at 400 pairs and lands at $+1.43$ and $+1.58$ on the same two banks. The gap is inside the intervals and is what a smaller bank plus a real integration should look like; both are quoted rather than the flattering one. Three limits travel with the table, two of which the next sections discharge. It is a *mirror match*, the arena in which a mechanism is most flattered (Section 6.3). The slope c is *fitted on this engine*, against this roster, under `us54` — one fitted constant, and the only one in the mechanism (Section 7). And the prototype touched *one hook*: nothing in the declare policy, the containment policy or the adaptive layer was changed.

6.3 The gauntlet: against opponents that also defuse

The fishlabs corpus supplies the warning that motivates this experiment: their structurally analogous inverted coordinate measured $+2.71$ against a weak opponent, $+0.55$ against a middling one and -0.99 against their strongest — a gain that existed only against opponents unable to punish it. Those three figures are recorded in this project's own concession document rather than in the cross-play one [1]. So the gate is whether defusal still pays when the opponent defuses too. Both arms face the *same fully-armed opponent* and only the measured team's appetite changes: arm A is `balanced(defuse 1)` against S(`defuse 1`), arm B `balanced(defuse 0)` against the same opponent,

Table 3: The gauntlet: both arms against the same fully-armed opponent, 250 duplicate pairs per cell [1]. Arm A against `balanced` is two identical policies on duplicate deals and must sum to zero by construction.

opponent style	arm A	arm B	**delta**
balanced	0.000	−1.636	$+\mathbf{1.64 \pm 0.40}$
blitz	−0.028	−1.744	$+\mathbf{1.72 \pm 0.45}$
punter	−0.228	−2.156	$+\mathbf{1.93 \pm 0.45}$
banker	+0.436	−1.112	$+\mathbf{1.55 \pm 0.55}$
turtle	+3.304	+1.724	$+\mathbf{1.58 \pm 0.56}$
hoarder	+1.300	−0.584	$+\mathbf{1.88 \pm 0.46}$
scout	+1.876	−0.088	$+\mathbf{1.96 \pm 0.51}$
ghost	+0.580	−0.504	$+\mathbf{1.08 \pm 0.52}$
archivist	+0.696	−0.916	$+\mathbf{1.61 \pm 0.53}$

250 duplicate pairs per cell on a third bank, reported as A minus B.

The gain survives at every style and is if anything larger than in the mirror match. Three features deserve reading separately. **Arm A against `balanced` is exactly** 0.000: two identical policies on duplicate deals must sum to zero by construction, and they do to the last digit — a free internal-validity check on the harness, and it passes. **Arm B is negative almost everywhere**: a team that does not defuse *loses* to one that does, by roughly the margin it would have gained by switching on, so the mechanism is contested rather than a coordination gain both sides collect, and declining to play it is a position rather than a neutrality. And **`ghost` is the narrowest cell** (+1.08), the style with the most suppressed information use and therefore the least to defuse with — an ordering that is mechanistic, and mild evidence that the effect is the stated one. This is not unconditional — every opponent here is still a FishAI policy, and the strongest *known* opponent to this engine need not be the strongest one — but the specific failure the corpus predicted did not occur.

7 The cross-engine replication

Section 6.3 answers "does it survive a stronger opponent" without answering "is this a fact about Canadian Fish or a fact about this roster". The second question needs a different engine, and there is one: FishAI's `decide` was run as a guest inside the **fishlabs** TypeScript engine, against *its* bots, with `defuse` the only knob moved [2].

7.1 What could not be played, stated first

The owner's request was for simulations against **SESTINA v1.0**, that project's frontier agent. **No game below is a game against SESTINA v1.0.** It is C++ and this machine has no C++ toolchain — `g++`, `clang++`, `cl`, `gcc`, `make` and `cmake` are all absent — so their binary cannot be produced here. What was played instead is their *TypeScript* lab engine and the eight bots shipped in it, which run under Node with no compiler. Their strongest TypeScript bot is FishBot v0.3; their lineage runs to v0.7, which is SESTINA v1.0, so the opponent here is roughly four releases behind their frontier [2]. One naming correction travels with that: the agent was briefly called "SISTINA", a first search returned nothing, and it momentarily looked as though it did not exist. It does, under the other spelling [19].

Note added September 2026. The compiler has since been installed and SESTINA v1.0 has been

Table 4: Defusal inside the fishlabs engine against the fishlabs bots, 300 duplicate pairs per opponent, `defuse` the only knob moved [2].

opponent	`defuse:` 1 net sets	`defuse:` 0 net sets	**paired delta**
`fishbot` (v0.3)	−1.333	−2.880	**+1.547 ± 0.517**
`fishbot_v02`	−0.933	−2.627	**+1.693 ± 0.555**
`lockout`	−0.573	−2.327	**+1.753 ± 0.533**
`detective`	−0.593	−2.300	**+1.707 ± 0.560**

played: their engine built from source, FishAI registered as a guest bot over their JSON-line protocol, 7,200 games, lost at 27.08% [18]. Nothing in this section is withdrawn, because none of it was a claim about what is possible in general — no game in *this* paper is a game against SESTINA v1.0, and the standing reported below is still the standing this paper measured, in the engine it measured it in. What the later match changes is how that standing should be read. It is not a proxy for the frontier: Section 7.3 finds FishAI one to seven points under even against their top four TypeScript bots, and against the agent four releases above those bots the deficit is 22.92 points of win rate, so the near-parity below does not extend upward. The defusal transfer of Table 4 is unaffected — it is a paired within-opponent delta, and the later match moved no knob it depends on.

7.2 The bridge, and the handicap it imposes

A bridge that quietly changes the game measures nothing, so the two rule sets were compared before any number was trusted. Deck, seats and teams, ask legality, turn flow and the wrong-declare outcome are all identical between the fishlabs TypeScript engine and `us54`, which makes the adapter a pure renaming and means FishAI's inference layer reads a log that means exactly what it means at home. One row differs: in their engine **only the turn-holder may declare**, where `us54` row 11 offers every seat the option in the declare window [2].

That difference handicaps FishAI, and it was quantified rather than waved at. Measured in FishAI's own engine over 300 `us54` games, 54.6% of its declares are made on its own turn — the channel the host provides — and **45.4**% off-turn, through a door their engine does not have. Some of those sets would simply be declared later on the seat's own turn, so 45.4% is an upper bound on the loss rather than an estimate of it; but the direction is unambiguous, and every FishAI win rate in the host engine is a **floor**. The adapter fell back to a default in 6 decisions out of roughly nine thousand (0.07%), always on the declare-window move that does not exist there, and results are bit-identical with and without it.

7.3 Standing, and then the result that matters

First, standing. At 150 games per cell in both orientations, FishAI at style `balanced` wins 42.67% against `fishbot` (v0.3), 45.33% against `detective`, 47.00% against `lockout` and 48.67% against `fishbot_v02`, and 80.67 to 99.67% against the other four bots in their lab [2]: competitive with their top four, one to seven points under even against them, while playing without the channel that carries 45% of its declares. Those four cells publish no intervals and Section 13 reclassifies them accordingly.

Table 4 is the strongest evidence in this project that defusal is a property of the game rather than of FishAI's roster. Four independent opponents, four intervals excluding zero — and the effect *size* transfers: +1.55 to +1.75 here against +1.50 and +1.65 on FishAI's own held-out banks and +1.08 to +1.96 against FishAI opponents that also defuse. The mechanism was derived, fitted and tuned entirely against FishAI's roster and had never seen any of these bots. Note what the table does *not* say: the `defuse: 1` column is still negative against all four. Defusal recovers about 1.6 sets of a deficit it does not close; FishAI loses to their top bots either way, by less with the mechanism on.

Since SESTINA is not among the opponents measured here, their reported evidence is cited as theirs [19, 2]: SESTINA v1.0 beats their pre-registered target by +3.33 pp $[+2.88, +3.78]$ over 48,000 games and the deployed v0.6 policy by +4.63 pp $[+4.19, +5.06]$ over the same 48,000. *Frontier margins in this game are small* — a seventh development cycle with a determinized search over a particle belief buys about three percentage points — which puts +1.6 sets per duplicate pair in perspective as a large effect by this game's standards.

Their README pairs those margins with three qualifications, and quoting the margins without them would overstate the frontier this paper measures itself against. They report that SESTINA costs **at least** 4.52× the non-searching blueprint. They publish a negative about their own flagship: it does *not* measurably beat a composite configuration assembled earlier in the same development cycle. And they decline the strongest claim outright, separating "strongest configuration this project has produced", which they record as established, from global standing, which they record as *"not established, not claimed"* — so there is no published basis, theirs or this paper's, for treating SESTINA as the strongest Canadian Fish agent in existence [19, 2].

7.4 `lockout`: the owner's rule, built independently, and doing well

The tension in this paper's central negative result is in their repository, and it is recorded here rather than buried. Their strategy table describes `lockout` as a turn-starvation specialist that ranks asks by posterior and then *charges a penalty for missing into a dangerous opponent* — the owner's off-limits rule, arrived at independently by someone else. And in their own round-robin (every fishlabs bot against every other, 100 games × 2 orientations per pairing) it ranks second of nine at 77.06% mean win rate against the field, behind `fishbot` v0.3's 82.69% and ahead of `fishbot_v02` (75.50%) and `detective` (75.31%) [2, 19].

This looks like a contradiction of Section 5 and it is not, for a reason that separates a strategy from a term. `lockout` is a whole strategy row — its own declaration threshold, its own risk parameter, its own targeting — that happens to *contain* a danger penalty. Its 77% says a strategy built that way is strong; it does not isolate the penalty. Section 5 is an ablation: one policy, one term added, everything else byte-identical, duplicate deals, plus an information-free null at matched magnitude. Only the ablation answers "does the danger penalty help".

A supporting argument offered here has been withdrawn. An earlier draft read the 34.1% ask accuracy recorded in the `lockout` cell as `lockout`'s own, and took it as this paper's mechanism visible from outside — a policy that avoids its best asks showing a low hit rate. That was a misread of a column that is FishAI's. Measured properly, both sides ask at almost exactly the same rate there (34.1% against 34.8%; and 49.8/50.6, 45.4/46.0, 43.5/44.4 in the other three cells), so the low rate is a property of the *matchup*, depressing both sides alike [2]. The inference is retracted rather than restated; what survives is the narrower observation that `lockout` produces the largest `defuse` delta of the four (+1.753), consistent with an opponent that publishes bases freely but establishing nothing about why. So the tension is left unresolved, which is the honest state, and only an ablation of `lockout`'s own coefficient inside their engine would settle it — which their lab does not expose as a knob.

7.5 Common failure points, theirs and this paper's

The owner's request named common failure points explicitly, and the host lab answers it most sharply through its own weakest serious bot. `kv_search` collapses on *declaration calibration*, not on play. It takes 30.25% against the field and 0.0% against each of the top three, and over 40 games against `fishbot` it takes 1.5 sets to 7.5 — but its ask accuracy in those games is 49.5% against 62.1%, a gap far too small to carry that result. Its declaration accuracy does carry it: **73.0**% against `fishbot`'s **95.3**%, from the lowest declaration threshold of any serious bot in their table (0.75) [2]. It is not being out-asked into oblivion; it is *gifting sets*. Under a rule set where any declare error awards the set to the opposing team — `us54` row 14, and their engine agrees (Section 7.2) — a 27% misdeclare rate is a losing position no amount of asking recovers. The lesson generalises beyond their bot: **under gift rules, declare calibration dominates ask quality.**

It also lands on this paper, which is the most self-critical thing the cross-play produced. FishAI's own declare accuracy against their bots is 90–94%, against `fishbot`'s 97%, and that gap is a plausible share of the one-to-seven points FishAI sits below their top four in Section 7.3 [2]. Its other visible weakness in this host is the declare channel of Section 7.2: without the off-turn window it must hold a located set until its own turn comes round, and a set held is a set an opponent may take a card out of first.

8 Signalling: refuted, and its observational support was reverse-causal

An ask publishes two facts to *teammates* as well as to opponents. The hypothesis is natural and its observational support is strong: sets that are declared correctly have been asked into far more often than sets declared wrongly. `scripts/probe-declare.mjs`, 200 games per style, every declare classified by whether the ask channel was still open — i.e. whether row 8 still permitted an ask at all — gives, for Balanced and Punter respectively: 1,468 and 1,469 declares with the channel open, **100.0**% and **99.7**% of them correct; 90 and 84 with it closed, **78.9**% and **75.0**% correct; and a mean of 11.70 / 11.80 prior asks into the set before a *correct* declare against **7.05** / **7.28** before a wrong one [1].

The intervention did not reproduce the association. A bonus for asking into sets the team cannot yet prove, weighted by proximity to channel-close, measured +0.013, +0.023, +0.067, −0.090, −0.203 at rising appetite over 300 pairs — and the one positive arm **failed to replicate**: 1,500 duplicate pairs on a disjoint bank gave 49.97%, -0.013 ± 0.033. The likely reading is that the correlation is reverse-causal: a set asked into a lot is a set several seats hold a basis in, and *that* is what makes it provable, so the ask count is a symptom of provability rather than a cause of it. A second version is also null — spending only the *free* choice, a tie-break among near-equal asks preferring the one that most reduces the team's allocation ambiguity at zero cost in hit probability, measured $+0.008 \pm 0.074$ and -0.010 ± 0.069 on two banks.

8.1 The structural objection, which is bigger than signalling

Two things this refutation does not rule out, and the first governs Section 9 as well.

A structural objection to the whole family. A linear ask score whose positive mass is hit-probability-driven cannot express a *deliberate miss*: the hit term is zero by construction and every remaining term is a penalty, so such an ask can never win an argmax at any weight. FishAI's score has exactly that shape — set p to 0 and 70 of the mass vanishes. So "signalling is inert when weak" may be a fact about the scoring function rather than about signalling; the measured *harm* at

Table 5: Stalling swept over trigger width and strength [1]. "Decisions changed" counts trigger positions at which the mechanism moved the argmax.

trigger width	strength	trigger positions	decisions changed	win rate	paired diff
2	1	332	16	50.00%	$+0.010 \pm 0.024$
6	1	1,952	514	49.67%	0.000 ± 0.052
3	20	552	440 (80%)	42.00%	$\mathbf{-0.42 \pm 0.12}$
6	20	1,471	1,330 (90%)	**33.33%**	$\mathbf{-1.14 \pm 0.19}$
6	100	1,421	1,329 (94%)	31.83%	-1.22 ± 0.19

high weight is real, but the inert regime is precisely what a structurally unselectable tactic looks like. Defusal escapes the trap only because a defusal ask *is a hit*, and concealment (Section 11) only because it moves the argmax between two hits.

The receiver side is untested. Everything above changes what the *sender* asks. A teammate that reads the log's row-6 and row-7 facts more aggressively when forming a declare plan is a different experiment, and has not been measured.

9 Stalling: refuted

The owner's third request was an endgame stall: decline to take the opponents' last card, to keep the ask channel open. Under `us54` there is no endgame phase — the rule set specifies `wholeTeamOut: 'declareWindow'` and the reducer simply moves the turn to the next seat with cards. What emptying the opponents does is end the **ask channel**, since row 8 requires a target holding cards, after which every remaining set must be declared on the log as it then stands. The strategic content of the request survives; the endgame framing does not [1, 7]. Measured incidence over 200 games: a team empties with sets unresolved in 41% of games, 72% of those caused by an ask the other side could have declined to make — but only **1.32** sets remain unresolved on average when it happens, the clinch at five sets making deep endgames rare by construction.

Table 5 is inert when weak, severely harmful when strong, and monotone — and not for want of alternatives: at trigger positions there are on average 23.15 legal asks across 2.71 distinct targets, and only 5.9% of positions offer a single target. The mechanism has room to act, and acting is bad. Refusing a card you can certainly take refuses free material *and* a retained turn, and what it buys is time on a channel whose product is worth very little — declares are already about 100% correct while the channel is open, and only about 1.32 sets remain when it closes. It is paying a certain card for insurance against a rare, small loss. Section 8.1's caveat applies here with more force: a stall is a deliberately worse ask, and the score cannot express one.

10 The turn handoff: built, and the shipped rule already wins

The owner also asked that a seat which runs out of cards on its own turn choose the teammate to pass to by reviewing the history — which teammate could best use the turn — rather than by hand size. The threat layer of Section 4 is exactly that estimator, pointed at a teammate instead of an opponent.

A win-rate A/B has no power at this decision, so the instrument is local: at every real pass decision, fork the game once per candidate teammate, force the pass there, and count the cards that teammate then actually takes before losing the turn — ground truth per candidate [1]. First, how often the decision exists at all, over 300 `us54` games: 11.3% of games reach `awaitPass`; of those

Table 6: The fork study: 4,000 games, 223 real pass decisions, 446 candidate rows [1]. "Picks best" is against the per-candidate ground truth the fork provides.

estimator	corr. with cards taken	mean cards captured	picks best
oracle	—	1.251	100%
`handSize` (shipped, `passTarget: 'most'`)	**−0.261**	**0.897**	**84.8%**
`reach` (the threat estimator)	**+0.352**	0.906	85.2%
hybrid (reach, hand size breaking ties)	—	0.892	84.3%

events 62.9% present a real choice (more than one teammate holding cards), 0.0% have no teammate holding cards, and 11.4% have the ask channel already dead — the owner's blind-declare case, which is 4 occurrences per 300 games. Mean unresolved sets at the pass: 1.74.

Table 6's correlations keep the signs of Section 4.2 — hand size negative, reach positive — and the *decision* is still unaffected, because **73.1%** of these decisions have no signal in them at all: every candidate takes the same number of cards, so no estimator can separate them. Among the roughly 60 that do carry signal, hand size and reach mostly agree. The shipped rule is already within 0.35 cards of the oracle and picks correctly 84.8% of the time.

Capability 5 is therefore **built and refuted as a change**, not skipped. **The estimator is right and the decision is empty**: +0.352 against −0.261 says the model is the better one, so this is a statement about how rarely `us54` asks the question, not a failure of the threat model. **Where it would matter is frozen**: the states in which this decision fires in 41% of games belong to the 48-card rule set, which this project holds byte-identical [8]. And **one premise of the request does not hold under `us54`** — "which teammate could best declare" is not a reason to pass, because row 11 gives every seat the declare option without the turn. The exception the owner identified is real and is the one case where it inverts: when the opponents are all out, the declare option sits on the turn-holder alone — the 11.4% of pass events above, or 4 occurrences per 300 games.

11 Concealment: a counter-weapon, not an improvement

The owner asked for a fourth mechanism after the first three were measured. When an opponent asks you for a card of set H, the natural reply is to ask into H — but that reply publishes your own row-6 basis in H. The proposal is to hold the card quietly instead, so opponents infer you have nothing there and skip you, until you can locate the rest of H and declare it.

11.1 The behaviour it targets, and why the mechanism is selectable at all

Before building a countermeasure, the behaviour was measured. After a seat is asked for a card of H, misses, and takes the turn under row 10, **88.1%** of its next asks go into H, against an availability-matched control of **21.4%** [3]. The pattern is real and large. **But the engine is not reciprocating; it is collecting certainties.** 99.1% of those replies are the argmax on hit probability and 60–94% are *certain hits*: the opponent's ask, combined with what the seat knew, located a card exactly, and the certainty bonus plus row 9 do the rest. Concealment therefore proposes to decline a certain hit, the arithmetic that killed stalling.

It nonetheless escapes Section 8.1's trap, because it is not a deliberate miss: it moves the argmax *between two hits* — the concealed ask and some other set's ask — and its charge is paid at most once per set rather than at every decision, since a second ask into H publishes nothing the first did not. So it is selectable at moderate weight, and measures non-zero where signalling and stalling measured

Table 7: Concealment added to defusal, against opponents that defuse [1]. The last row is the cell that decides what the mechanism is; Section 13 reclassifies it as unresolved rather than zero.

arm	N	paired set-difference
byte-exact control	—	$\mathbf{0.0000 \pm 0.0000}$
held-out bank A	800	$+0.8875 \pm 0.2443$
held-out bank B	800	$+0.8325 \pm 0.2378$
headline	600	$+\mathbf{0.9683 \pm 0.2744}$
information-free null of the same shape and magnitude	—	$-\mathbf{0.7200 \pm 0.2715}$
against opponents that do NOT defuse	600	$-\mathbf{0.1483 \pm 0.2571}$

flat. It is charged as $a_c \cdot w_{\text{hit}} \cdot c \cdot \text{mine}(H)/(1+E)$ — deliberately the same conversion Equation (3) uses, so that the two mechanisms arbitrate exactly rather than by accident. Setting the two against each other, defusal wins iff $a_d \cdot p \cdot \text{prey}(H) > a_c \cdot \text{mine}(H)$, and at both appetites 1 that is simply

$$p \;>\; \frac{\text{mine}(H)}{\text{prey}(H)}: \tag{4}$$

take the card back when it is likely enough relative to how much of the set you would be advertising. Measured over 13,622 ask decisions the two mechanisms collide on 28.6%, and defusal takes 79.8% of the collisions. They compose; neither cannibalises the other.

11.2 The measurement, including the part that decides it

Table 7 carries a dose-response in how well the opponent can read the leak concealment hides: +0.97 against a hard-skill opponent, +0.31 against medium, +0.11 against one without constraints. **The entire effect is denial of defusal** — not a general improvement but a counter to one specific mechanism.

A correction, printed where the claim was. This section used to end with *"removing only the opponent's defusal drops the marginal value to* +0.157*"*. That is withdrawn. It contradicts the table's own last row in sign, and it does not reproduce: measured directly, conceal-only against plain is -0.2883 ± 0.2640, and as a difference-of-differences on a common bank -0.3067. Both readings come out negative, and where the +0.157 came from could not be reconstructed, so it is retracted rather than reinterpreted [1].

And the row that decides the mechanism is below the instrument's resolution. -0.1483 ± 0.2571 at $N = 600$ sits under a minimum detectable effect of about 0.38 sets at that cell's own per-pair SD (Section 13), so concealment against a non-defusing opponent is **not resolved here**: anything from a real loss of about a third of a set to a small gain is consistent with it. What *is* resolved, at 4,000 pairs per cell, is that conceal-only is the weakest of the four concession arms outright (Section 12).

11.3 Verdict, and the correction it forces on defusal

Not shipped on. `conceal` is landed as an optional style field defaulting to absent, so every roster vector and every shipped difficulty tier is byte-identical, and the tiers carry `defuse: 0` and so have no customer for a counter to defusal.

This section used to force a correction on Section 6, printed in the body as a warning that *"defusal's* +1.50 *is an over-estimate against a concealing opponent"* which would take "roughly one set per duplicate pair back" — on the grounds that neither the gauntlet (Table 3) nor the cross-engine

Table 8: All six edges of the 2×2 through one code path, 4,000 duplicate pairs per edge, with only the arm definitions changing; then reproduced by an independent agent on six entirely fresh banks [1]. Arms: **P** plain, **D** defuse-only, **C** conceal-only, **DC** both.

edge	first run	independent replication
D vs P	$+1.6225 \pm 0.1016$	$+1.6595 \pm 0.1022$
C vs D	$\mathbf{-0.6797 \pm 0.1030}$	$\mathbf{-0.5102 \pm 0.1051}$
P vs C	$+0.1457 \pm 0.1030$	$+0.1742 \pm 0.1030$
DC vs D	$+0.9630 \pm 0.1069$	$+0.8207 \pm 0.1070$
P vs DC	$\mathbf{-1.2985 \pm 0.1036}$	$\mathbf{-1.3155 \pm 0.1046}$
DC vs C	$+0.3690 \pm 0.0987$	$+0.3202 \pm 0.0987$
byte-exact control	0.0000 ± 0.0000	0.0000 ± 0.0000

replication (Table 4) ever put defusal in front of an opponent that conceals. **That gap has since been measured, and it closes the opposite way** [1]. Table 8's C vs D edge is precisely the missing cell, at 4,000 duplicate pairs per edge on two independent bank sets: a conceal-only opponent does not take a set off a defusing one, it *loses* to it, by -0.6797 ± 0.1030 and -0.5102 ± 0.1051 — about half a set per pair, in defusal's favour, on both banks.

The worry was the right shape aimed at the wrong target. The figure near 0.9 belongs to a different comparison entirely: concealment *added to* defusal on the same side, the DC vs D edge at $+0.9630$ and $+0.8207$ (Table 8). That prices what a defusing team gains by concealing as well — a fact about its own configuration — and says nothing about what a concealing opponent takes off it. Defusal's headline stands as measured.

12 The triangle, refuted, and what replaced it

This project published a claim about its own tactic space: that defusal and concealment made it *intransitive*, and that the resulting cycle was the reason an arbitration layer over per-decision tactics would pay. Measured properly, there is no cycle, and the retraction and its replacement are reported here as a first-class result.

There is no cycle under either reading of "the concealing corner", and the loop is broken by a wrong-signed leg rather than a straddling one — so no sample size rescues it. Read the corner as conceal-only and the second leg reverses: concealment does not beat defusal, it loses to it by about half a set (9.5–12.9 SE from zero). Read it as defuse-plus-conceal — which is what Table 7's +0.97 actually measured, concealment *added to* defusal against defusing opponents — and the second leg holds but the third reverses instead: DC beats P by 1.3 sets (12.5–24.6 SE). The full 4×4 is a **strict transitive total order,** $DC > D > P > C$, every edge the same sign on both bank sets. Concealment is an increment on defusal, never a substitute for it, and **conceal-only is the worst of the four arms** — worse than plain.

Two further corrections came out of the replication rather than out of the run. The triangle table used to quote defusal at $+1.92 \pm 0.31$ from a 400-pair common-baseline bank; at 4,000 pairs it is +1.62 / +1.66, ordinary sampling variation — and the first explanation offered for that gap, that the mirror harness lacks `pickAsk`'s certain-hit gate, was tested by running the ungated harness and *refuted* ($+1.6850 \pm 0.3358$, indistinguishable from the gated value). And the claim that the published triangle mixed two definitions of the concealing corner is **not** established: `conceal.ts`'s own header points the other way, and the two readings cannot be told apart at this resolution.

Table 9: Which numbers in this paper sit below their own cell's floor [1, 17]. The MDE depends on *that cell's own* SD; this is a record of specific verdicts, not a threshold to apply by eye.

result	reported	N	its floor	verdict
Table 7, conceal vs non-defusers	-0.1483 ± 0.2571	600	≈ 0.38	**below floor — not resolved**
The contained-pass aim, "inert"	-0.045 ± 0.130	—	—	**below floor — a non-measurement**
§7.3, all four headline cells	42.7–48.7%, no CIs	150	1.05–1.13	**all four below floor**
Table 2, defusal shipped	$+1.43$ / $+1.58 \pm 0.32$	400	≈ 0.47	resolved, 3× the floor
Table 8, the six edges	± 0.10 half-widths	4000	≈ 0.15	resolved

12.1 What survives, and why it is stronger than the cycle was

The old argument ran: the v1.0 adaptive layer degenerates because its counter table has a dominant row [12]; a defuse/conceal pair has no dominant row; therefore reading the opponent starts to pay. The premise was wrong. The conclusion is right anyway, for a plainer reason — **the column argmax is opponent-dependent.** Measured on common banks, paired on the deal twice over, the best response against a *plain* opponent is D by $+0.2870 \pm 0.1392$ over DC; against a *conceal-only* opponent it is again D, by $+0.1462 \pm 0.1437$ over DC; and against a *defuse-only* opponent it is DC, by $+0.9152 \pm 0.1353$ over D [1].

The best response flips on one bit of the opponent's configuration — *does it defuse?* — and knowing that bit is worth up to +0.29 sets per duplicate pair against a non-defusing opponent, which is not a hypothetical opponent: every shipped tier carries `defuse: 0`. It is Section 11.2's own thesis restated as a policy — conceal only against an opponent that defuses — and it makes the follow-on work concrete and *smaller* than a portfolio. The v1.0 degeneracy verdict does need revisiting, since degeneracy there is defined as "the read never changes the argmax" and here the read changes the argmax; but the table it needs is one binary coordinate, not a tenth style [12, 5]. The arbitration layer is still the larger question and still unbuilt.

13 The detection floor, and the two nulls it reclassifies

Every headline in this paper comes from one instrument: K duplicate pairs of arm X against arm Y under `us54`, reported as the mean paired set-difference ± 1.96 SE. Until it was characterised, every published null was ambiguous between "no effect" and "no resolution". That characterisation is a sibling result in this batch [17]; what matters here is what it does to this paper's own numbers. The per-pair SD is 3.15–3.44 sets for arms that differ at every decision (replication: 3.18–3.44), giving an 80%-power minimum detectable effect of 0.76 at $N = 150$, 0.54 at 300, 0.47 at 400, 0.38 at 600, 0.33 at 800, 0.21 at 2,000 and 0.14 at 4,300 pairs.

The power model is the solid part: across twelve (planted-effect, N) cells the observed detection rate sat within about one binomial SE of normal-theory prediction in both runs. Two parts did not replicate, and both cut against the first run's conclusions — the empirical floor at $N = 400$ (84.7% detection of a planted 0.393, then 75.0%; pooled 81.7%, so the floor is somewhere in 0.36–0.50) and interval coverage at $N = 400$ (6.1% false positives, then 10.8%; pooled 8.0% $[5.2\%, 11.7\%]$, excluding the nominal 5%). At $N = 800$ both runs are nominal [17].

Table 9 is the discipline applied to this project's own work. **Two of this paper's nulls are reclassified as non-measurements**: the contained-pass aim called "inert" in Section 14, and the non-defuser cell of Table 7. Neither is a statement about the game; both are statements about this instrument's resolution. A third reclassification lands on the cross-engine standing figures: at 150 games per cell the per-opponent ranking of Section 7.3 is not resolved, which is why Section 7 rests on Table 4's 300 properly paired duplicate pairs and not on it.

One trap is worth naming, because the first floor run fell into it: the MDE depends on that cell's own SD, and the SD is not constant across the repository — the turtle cell of Table 2 measures 3.645, and the cross-engine harness's own published half-widths imply it runs at 4.57–4.95, some 38–49% higher [17]. So one SD substituted across all conditions understates the floor exactly where the cells are smallest. Everything this paper ships a mechanism on — Tables 2, 3 and 8 — clears its own floor by 3× or more.

14 Two defects in the shipped engine, found on the way

1. `missTarget: 'fewest'` is backwards. Its trace prose tells the user that the smallest hand makes the surrendered turn "worth least"; the measurement says the opposite in every phase bucket (Section 4.2), and the fork study of Section 10 reproduces the sign a third time at -0.261. The same proxy appears in `valueContainedPass`'s n_t term and in the contained-book policy's aiming rule [16].

The fix was built where it is soundest, and measured inert. A contained-book turn-pass is a *guaranteed miss* by construction, so Section 5.2's danger/opportunity confound cannot arise there: it is the one place in the engine where "concede to the least dangerous seat" is exactly right. Replacing both the aim and the cost model with the threat estimate measured -0.045 ± 0.130 over 400 duplicate pairs, and the diagnostic says why: over 25,654 ask decisions, more than one opponent still holds cards 99.1% of the time, the contained pass fires at all on **1.06**%, and it fires with the two aims disagreeing on **0.11**%. Published reach is too sparse there: most opponents have shown no basis at all, so every candidate prices the same and the aiming gain collapses.

The change was reverted rather than shipped off by default, on the argument this project applies to its own knob inventory — a knob that is listed, swept and reported while being inert contaminates the measured vector, and one of those is already one too many. Section 13 then reclassifies the -0.045 itself: it is below the floor, so "inert" overstates what was measured, and the revert stands on the sparsity diagnostic and the contamination argument rather than on the null.

2. `style.signalling` is unreachable under `us54`. The signalling ask is gated inside the planner path, which a `us54` game enters only for `phase === 'endgame'` — a phase that rule set never produces. So the knob is listed, swept and reported, and buys a `us54` seat nothing at all except through the contained-pass information price, where it zeroes one term. **A dead knob in a measured vector is a contaminated comparison**, the same argument that decided the revert above. This project has published that failure mode once before, for a different axis [15]; this is the second instance in the same roster.

15 Capability 4: switching actions instead of playstyles

The owner asked for an adaptive model that, "instead of switching between playstyles, just switches between actions and strategies, to allow for more flexible playing". **That is the right architecture, and it is now measured rather than argued.** The two layers price against each other in the same unit: the v1.0 adaptive layer, which switches the whole style once per phase, is worth $+0.13 \pm 0.06$ sets per duplicate pair over a bare Balanced team [12, 5, 6]; the style knob itself, chosen well, is worth nothing, because it is a constant (a ± 0.27 spread); and the per-decision terms (`containedPass`, `defuse`), which switch the *action* at every decision from the position, are worth $+1.43$ and $+1.58 \pm 0.32$ (Table 2).

The per-decision layer is worth roughly eleven to twelve times the style-switching layer, and the style-switching layer is worth less than simply naming a better style would have been — the owner's point, quantified. One arithmetic correction belongs here rather than in a footnote: the project's own summary table records this ratio as "about 15×", and against the numbers this paper reports it is not, since $1.43/0.13 = 11.0$ and $1.58/0.13 = 12.2$. No source records what the larger figure was computed from. The plausible reconstruction — offered as one, not as a provenance — is that it is the ratio against the defusal estimate this paper has withdrawn, Section 12's retired +1.92, since $1.92/0.13 = 14.8$. What is not in doubt is the ratio of the numbers actually carried, which is eleven to twelve.

There is a structural reason the style layer cannot do better, and it is a theorem rather than a measurement: best-response selection over the committed counter table provably degenerates to always-Punter, because one row weakly dominates every column and the expected payoff is linear in the opponent posterior [12, 5]. A layer that reads the *position* — has this opponent published a basis in a set my team can locate cards of? is this set contained? is the ask channel about to close? — is choosing from the legal move list instead, every turn, and every mechanism in this paper is one predicate over that list. So what is shipped is a *step* of capability 4, and the arbitration layer over it is worth building only now that there is more than one tactic to arbitrate: of the four originally proposed three measured harmful or inert, and Section 12.1 prices the arbitration between the two survivors at one bit.

16 Threats to validity

1. **One fitted constant, fitted here.** The cards-per-prey-card slope of Equation (1) is measured on this engine, against this roster, under `us54`. Table 4 is evidence that the *mechanism* transfers, not that the constant is optimal elsewhere; no re-fit was attempted inside the host engine, because that would have burned it as a holdout [2].

2. **A stated K is missing.** Several variants of the defusal term were scored before the winner was picked — per-set against per-seat, five appetites, two gate designs — and the selection haircut is unpriced. The held-out banks and the cross-engine replication stand in for it; neither substitutes for the arithmetic.

3. **The threat model is an under-approximation by construction.** It reads no hand but the viewer's own, so it is silent about every basis a seat has not published and is worst early. That property is what makes avoidance unsound (Section 5.2); it also caps how much of the true threat any of these mechanisms can act on.

4. **The opponents are all policies of two engines**, and both populations are closed. The gauntlet's opponents are FishAI policies; the cross-engine opponents are four bots roughly four releases behind their own project's frontier, played without the declare channel that carries 45.4% of FishAI's declares (Section 7.2). Neither population contained a *concealing* opponent — but that particular gap is closed rather than open: Table 8's C vs D edge measures defusal against a conceal-only opponent at 4,000 pairs on two bank sets and puts defusal ahead by about half a set (Section 11.3).

5. **A channel the host engine has and FishAI does not was switched off.** Their engine supports `psychologicalTells`. FishAI has no notion of it, so it was left **off** for every cell in Section 7, on the grounds that leaving it on would have measured something other than

card play [2]. That is the right call for isolating the mechanism, and it is also a bound on the standing figures: their bots are not being played with everything the host gives them.

6. **Resolution, and the harnesses.** Section 13 is the honest boundary of every null above, and a null not named in Table 9 has not been individually checked against its own cell's SD. The measurements come from scratch harnesses under `scripts/probe-*.mjs`: not typechecked, no committed artifact, not digest-bound, unlike the lab's tournament artifacts [9, 10]. They exist so every table above can be regenerated, and that is the strongest claim available for them.

17 What is not settled

The residue is kept in the body, because a paper that shipped a mechanism and hid its open items would be misreporting the work.

- **The style matrix is invalidated.** The roster now carries `defuse: 1`, so the committed style matrix no longer describes the shipped roster and must be re-run and re-pinned [8].
- **A pre-existing defect in the committed search numbers**, found while adding the new knobs to the ladder: the candidate budget was the literal 60 against a 75-rung ladder, so the search evaluated a *prefix* and abandoned the tail — a prune selected by the order the fields happen to be typed in. Replaying the old ladder along each style's committed accepted moves reproduces the artifact's candidate counts exactly and shows what the prefix cost: `banker` never evaluated `minHandSize=3`, and `turtle` never evaluated `minHandSize=1` or `minHandSize=3` — three rungs across two of nine searches. The budget is now the ladder's own length. The exploit cache does not see any of this, because it keys on an engine hash that walks the engine tree only and neither the ladder nor the budget lives there, so a cached re-run re-emits the pre-fix numbers.
- **The concealment half of the certain-hit gate is unpinned.** `pickAsk` documents the gate as zeroing *both* terms; the gate test pins only defusal, because in its fixture the concealing seat has already published a basis in both sets it can ask into and the charge is exactly zero on both asks — asserted explicitly, so the test cannot be misread as coverage it does not have. A mutation leaving that charge live under a certain hit is invisible to the suite, and the gate's own firing rate is unpriced by any ladder (Section 6.1).
- **Sign-flip coverage was over-claimed and then corrected.** The claim was that both concession terms pass their suites with the sign reversed. Measured by reversing each of the three sign errors available: negating the defusal bonus fails 2 of 25 existing assertions, negating the concealment charge fails 3 of 25, and reversing the *application* in `pickAsk` — writing charge − bonus for bonus − charge — **passes both suites entirely**. One third of the claim was true and it was the important third: every assertion called the exported functions directly, and nothing pinned what `pickAsk` did with the numbers they returned. Two decision-level tests now close it.
- **Four seams are held by hand.** No test pins the concession layer's turn-yield equal to the contained-book policy's E, though both file headers describe them as deliberately duplicated; the tabled-licence export has no caller; the dynamic test of the public-view boundary runs the three shipped tiers, which carry `defuse: 0`, so no code in the threat, defusal or concealment files executes under it; and the bounded arm's licence scan retires licences against budgeted

knowledge while the ask history it scans is not itself budgeted, a fix that would change every committed v1.5 number [13, 6].

- **A true match against SESTINA v1.0 was possible, was blocked only by a missing compiler, and has since been run** (Section 7.1). Installing a C++ toolchain and speaking their documented JSON-line protocol made it a day's work, as predicted; the result is 27.08% over 7,200 games and is reported separately [18, 19]. What this item did not anticipate is that the first attempt returned 24.22% and was wrong: the adapter translated `us54`'s compulsion to declare into a host that does not impose it, and the mirror-cell control could not catch it because a mirror cell is forced to 50% by construction. The number was withdrawn rather than corrected. The remaining future work is therefore not the match but the 16.4 points of it that no mechanism yet explains.

18 Conclusion

The request that started this work was for a rule that refuses to ask dangerous opponents. The rule was built exactly as asked, measured against a byte-exact control on duplicate deals, and it loses — up to 11.7 points of win rate, monotonically worse as it is driven harder, while an information-free perturbation of the same magnitude is free. The threat model underneath the request is nonetheless correct, and better than what the engine already used for the same job: the shipped hand-size proxy correlates −0.147 with what a conceded seat actually takes, the licence estimator +0.421.

What makes the prescription wrong is one structural fact. Row 6 permits an ask only into a set the asker holds a card of, so the seat that is dangerous to you is the seat holding cards of the sets you are invested in — which is the seat most likely to hold the card you want. Threat and opportunity are the same measurement, corr = +0.249, and avoidance trades away your own best asks to buy safety that a public-log threat model can never certify anyway, since an absence of published reach is not evidence of absent reach. Invert it and the same model prescribes the opposite move: ask *into* the set the dangerous seat has published a basis in, take the card its reach rests on, and keep the turn under row 9. That is worth +1.43 and +1.58 sets per duplicate pair on held-out banks, is positive at all nine roster styles, survives opponents that defuse back at +1.08 to +1.96, and — the result the project would keep if it could keep only one — transfers at the same magnitude, +1.55 to +1.75, into a third party's engine against a third party's bots with one knob moved.

The rest of the ledger is negative and is reported at the same weight: signalling flat and unreplicated, stalling harmful once strong enough to act, the turn-handoff request beaten by the rule already shipped, concealment a counter-weapon rather than an improvement. Three items in it are this project correcting itself. The intransitive triangle it published about its own tactic space does not exist, and the opponent-dependent argmax worth one bit that replaced it is the better result. A fix for the engine's wrong-signed concession proxy was built where it is provably sound, measured below the instrument's floor, and *reverted rather than shipped off by default*, because a listed-but-inert knob contaminates every vector it appears in. And the certain-hit gate exists because an adversarial review found the new credit could displace a certain hit, which it then did nine times in a sweep against zero for the control.

Two of the nulls above are not zeros. The detection floor [17] reclassifies concealment's non-defuser cell and the contained-pass aim as *non-measurements* — this project's discipline applied to its own published numbers rather than to someone else's. Where a result is below the instrument's resolution, the honest report is the resolution.

Reproducibility. The engine is deterministic and one seed reproduces a byte-identical game

[10]; every table above is regenerated by a named probe under `scripts/probe-*.mjs` [1]. The cross-engine harness lives outside the repository, beside the clone it drives: that clone carries no licence file, nothing from it is copied into this project, and their findings are cited as theirs [2, 19].

References

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[11] A. Hsieh. FishAI v0.5: measuring play style without skill in a 54-card Literature variant. FishAI project; companion paper, 2026.

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[19] `dylann4500`. FishLab — a Canadian Fish engine, bot family and evaluation lab. GitHub repository, `https://github.com/dylann4500/FishLab`. The C++ SESTINA v1.0 agent and its reported margins, the TypeScript lab engine and its eight bots, the `lockout` danger-penalty strategy, and the JSON-line cross-play protocol. The repository carries no licence file; nothing from it is reproduced here.

[20] J. McLeod (ed.). Literature (Canadian Fish). Pagat card-game rules archive, `https://www.pagat.com/quartet/literature.html`. Folklore source for the common core of both rule sets this project supports.